Overdrive GP

A 3D, Formula One-style racing game set in a neon night world. It is a **web app** that runs in the browser on phone and computer, with a feel **between arcade and realistic**.

> Status: **planning only.** Nothing is built yet. Items marked *(suggested)* are drafts that have not been confirmed.

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1. Core Decisions

Area	Decision
Platform	Web app, playable on phone (touch) and computer (keyboard)
Style	Neon night race: dark tracks, glowing edges, light trails, reflective wet roads
Feel	Between arcade and realistic
Grid	6 teams, 12 drivers (2 per team) in single-player
Start	Player chooses team and driver (quick race)
Accounts	Sign in with Google, Apple, or email. Game data is linked to the account.
Guests	Can play quick races without signing in (progress not saved)
Not included	Battery management, fuel management, story mode

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2. Teams and Drivers

Each team has a neon color identity *(colors confirmed)*.

Team	Color	Driver	Personality (draft, approved)	Strong	Weak
Apex Velocity	Hot red	Jace Sterling	Confident golden boy	Qualifying pace	Composure under pressure

		Kaelen Vance	Cold, ruthless rival	Overtaking	Tire care
Vanguard Corse	Electric blue	Marco Rossi	Passionate showman	Slipstream, starts	Consistency
		Elena Moretti	Calm tactician	Tire care, long races	Risk-taking
Meridian Strategy	Cyan	Arthur Vance	Safe veteran	Consistency	Overtaking
		Silas Thorne	Quiet genius	Wet weather	Slow starts
Kinetic Strike	Orange	Jaxson Holt	Hot-headed daredevil	Speed after being passed	Crashes
		Renna Cross	Fearless, unpredictable	Fog and wet weather	Composure
Zenith Precision	Violet	Leo Ashford	Perfectionist	Qualifying, clean lines	Chaotic races
		Freya Bennett	Analytical rookie	Adapts during a race	Low early confidence
Obsidian Drift	Lime on black	Silas Hale	Moody lone wolf	Night overtaking	Teamwork
		Kai Mercer	Laid-back risk-taker	Comeback speed	Early-race focus

Story hook: Kaelen Vance and Arthur Vance share a surname and race for rival teams. There are also two drivers named Silas (Thorne and Hale).

Driver stats (set by personality)

Overtaking, tire care, wet-weather skill, consistency, qualifying pace, composure under pressure.

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3. Create Your Own Player and Team

- Players can create their own **driver** and **team**.
- A created team **replaces** one existing team, and a created driver replaces an existing driver, so the grid size stays the same.
- When a team is replaced, **one of its drivers can stay** as the new team's teammate.
- A "restore original" option brings the default team and drivers back.
- **Player options:** name, nationality, number, personality, helmet look, optional bonus points with a matching weakness.
- **Team options:** name, abbreviation, neon color pair, glow style, logo, livery pattern, car number style.
- A name filter blocks inappropriate names.

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4. Game Modes

Mode	Practice	Qualifying	Race	Account needed
Quick Race	No	Optional	Yes	No (guest ok)
Career	Yes (test upgrades)	Yes	Yes	Yes
Multiplayer	No	Yes	Yes	Yes

Career mode

- Racing progression only, **no story**.
- Start at the **bottom team** and work up.
- Seasons across the 10 tracks, with drivers' and teams' championships.
- Contract offers from better teams after strong seasons.
- Objectives and sponsor goals, such as "finish top 8".
- Rivalries driven by driver personalities.
- Free practice lets you test upgrades before buying.

Multiplayer

- Online, joined with a **room code** (no friends list needed).
- Up to **12 players**, each with their own created driver and team (one driver per team).
- Qualifying and main race only.
- Empty slots stay empty (no AI fill).
- Equal cars, so upgrades are off or fixed.

- Teams need distinct colors and numbers.
- Host sets track and weather. A dropped player can rejoin with the same code.

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5. Tracks (10 fictional night circuits)

Each track has its own pit lane, obstacles, and hazards. Lap counts are based on the track.

#	Track	Setting	Hazards	Pit lane	Laps *(draft)*
1	Neon Metropolis Circuit	City	Hairpin under a bridge, flickering tunnel	Main straight, low time loss	8
2	Harborlight Street GP	Coastal	Wet dock plates, crosswinds	Narrow, behind harbor wall	8
3	Mirage Dunes Raceway	Desert	Drifting sand, blind crests	Wide, easy entry	6
4	Skyline Loop	Rooftops	No runoff, moving barriers	Lower level, ramp access	10
5	Glacier Run	Frozen mountain	Ice patches, low visibility	Standard, cold tires matter	7
6	Voltage Park	Power plant	Timed laser gates, chicanes	Cuts through the circuit	9
7	Tidal Pier Circuit	Over water	Spray, narrow pier	Short, tight garages	10
8	Obsidian Canyon	Canyon	Rockfall, fast sweepers	Longest pit lane	5
9	Lumen Forest GP	Glowing forest	Branches, fog patches	Hidden among trees	6
10	Apex Infinity Speedway	High-speed ring	Banked turns, slipstream battles	Fastest pit lane	8

Target race time is roughly 10 to 13 minutes. Good starting tracks to build: **#1 or #10**.

6. Tires and Pit Stops

Tire wear is required in every race mode.

- **Compounds:** soft (fast, wears quickly), medium (balanced), hard (slow, lasts), intermediate (light rain), wet (heavy rain).
- Wear grows with hard braking, sliding, aggressive cornering, kerbs, and off-track driving.
- Each track has its own tire load. Heat and cold change how tires behave.
- HUD tire icon: green, yellow, red. Grip drops as tires wear.
- The race engineer warns you before tires run out.
- Driver personality changes tire care.
- **Setting:** wear rate (low, normal, high). In multiplayer the host sets it.

Pit stops

- Pit lane speed limit, with a time penalty for speeding.
- Quick tap-timing mini-game, with crew speed bonuses.
- Pit lane length sets the time cost on each track.
- "Pit this lap" button when near the pits.

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7. Weather, Crashes, and Flags

- **Weather:** clear, cloudy, light rain, heavy rain, fog. Rain can start or stop mid-race. HUD shows a forecast.
- **Wet effects:** less grip, puddles, reflective neon roads. Some drivers handle the wet better.
- **Crashes:** contact slows or spins cars. Damage is **visual only** *(assumed, not confirmed)*. After a spin the game places you back on track.
- **Penalties:** time penalties for causing a crash, cutting a corner, or pit lane speeding, with a clear message each time.
- **Flags:** yellow (no overtaking in the zone), safety car (pack bunches up, pit strategy changes), red (severe blockage).

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8. AI and Difficulty

- **Difficulty levels:** Rookie, Pro, Legend (affects AI speed, mistakes, aggression, and help).
- AI follows the same physics as the player, with no cheating speed bonuses.
- Each AI driver uses its personality and stats, pits by tire wear and weather, and reacts to rain.
- Drivers who clash with you become more aggressive toward you.
- **Rubber-banding is a setting** (off or light). *(Suggested defaults: off on Pro and Legend, light on Rookie.)*

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9. Car Upgrades and Money (Career only)

Part	Effect
Engine	Top speed and acceleration
Aero	Cornering grip and slipstream
Brakes	Shorter stops, less tire wear
Tires and suspension	Grip on kerbs and in the wet
Cooling	Push harder for longer
Pit crew	Faster stops

- 5 levels per part, with rising cost.
- Money comes from race finishes, sponsor goals, and clean-race bonuses.
- Money goes to upgrades and contract fees.
- AI teams also upgrade over seasons. A cap on upgrade points prevents an unbeatable car.
- Practice lets you test parts before buying.

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10. Controls

Phone

- Left: steer left, steer right.
- Right: gas, brake, and a boost button above them.

- Top: pause, camera switch, rear view.
- "Pit this lap" toggle near the pits.
- Optional tilt steering and resizable, movable buttons.

Computer

- Arrow keys or WASD to drive.
- Shift or Space: boost. C: camera. P: pit request. Esc: pause. Hold R: rear view.

Main menu: Quick Race, Career (Continue, New Career), Multiplayer (Create Room, Join Room), Create Team and Player, Settings, Sign In.

Race weekend buttons: start practice, test an upgrade, skip to qualifying, and "Ready" in multiplayer.

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11. Settings and Accessibility

- Difficulty, rubber-banding, wear rate (host controls in multiplayer).
- Assists: braking guide, steering help, racing line (each switchable).
- Glow level (low or full), reduced flashing, colorblind-safe colors.
- Button size and opacity, vibration, tilt steering.
- Camera view and sound volume.
- A short driving school teaches steering, braking, tires, and pit stops.

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12. Accounts and Data

Feature	Guest	Signed in
Quick race	Yes	Yes
Career	No	Yes
Custom players and teams	No	Yes
Multiplayer	No	Yes
Best laps and ghosts	This session only	Saved
Settings	Browser only	Synced across devices

- Saved to the account: career progress, upgrades, custom teams and players, settings, best laps, multiplayer name and stats.
- Needs a backend (server and database), which multiplayer also uses.
- Store only what is needed, give a privacy note, and allow account and data deletion.
- Use a trusted sign-in service rather than anything custom.

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13. Technical Notes (suggested)

- 3D in the browser with a standard web 3D library (such as three.js).
- Keep a low-glow mode and limit visible cars for phone performance.
- Audio starts after the first tap, which the menu screen handles.
- Hosting on a free service is enough to start.
- Multiplayer is the largest piece of work, so build it last.

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14. Suggested Build Order

1. A drivable car on one track (#1 or #10), working on phone and computer
2. Lap timer, countdown, tires, and basic pit stops
3. AI opponents and difficulty
4. Menus, sign-in, and cloud saves
5. More tracks and weather
6. Career mode, upgrades, and custom players and teams
7. Multiplayer with room codes
8. Sound, music, and polish

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15. Open Questions

- Should the multiplayer host be able to change race length (short, standard, long) or should laps always be fixed per track?
- Should crashes cost repair money in career mode, or stay visual only?
- Will younger players be expected? If so, plan age checks and extra privacy rules.
- Final lap counts and track lengths need tuning once tracks exist.
- Which features belong in version 1 and which wait for later?

- Sound, music, and commentary are not yet planned in detail.